

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Level 1

Teaching Guide

Level: Level 1

Class: Class 6

Duration: June - January (8 months)

Period: 40 minutes

ICT Teaching Resources Portal

Table of Contents

Curriculum Overview	3
Themes Covered	3
Pedagogical Approach	3
5E Model Implementation	3
International Pedagogies Used	3
Lesson Plan Structure (40 Minutes)	3
Assessment Framework	3
Assessment Components (Per Class)	3
Resources Required	4
Hardware	4
Software (Free/Open Source)	4
Dormitory Tasks (Zero Lab Access)	4
Capstone Projects	4
Project 1: School Environment Monitor	4
Project 2: Hostel Store Inventory	4

Curriculum Overview

ICT Level 1 is designed for Class 6 students in KREIS residential schools. The curriculum follows the 5E instructional model and integrates Singapore's Concrete-Pictorial-Abstract approach for effective learning.

Themes Covered

Chapter	Theme	Periods	Key Topics
1	Internet and ICT Environment	12	Introduction to ICT, Internet basics,
2	Graphics in ICT	12	Web browsers, GPS Painting/Drawing, Graphics
3	Audio & Visual Communication	12	software, Symmetric designs Communication types, Audio/
4	Data Representation & Processing	12	Video conferencing, Audacity Number systems, Data
5	Programming	12	representation, Basic processing Scratch programming, Basic blocks,
6	Software Applications	12	Simple animations, Introduction to software, Types of software

Pedagogical Approach

5E Model Implementation

Phase	Duration	Activities
Engage	5-7 min	Real-world connection, regional metaphors,
Explore	10-12 min	questions
Explain	10-12 min	Hands-on activity, Driver-Navigator pairs, Lab work
Elaborate	8-10 min	Live coding, Concept introduction, C-P-A approach
Evaluate	5-8 min	Extended activity, Parson's problems, Application Formative check, Exit tickets, Self-assessment

International Pedagogies Used

- **Singapore C-P-A:** Concrete manipulatives → Visual blocks → Abstract code
- **UK NCCE:** Unplugged Computing, Live Coding, Parson's Problems
- **Driver-Navigator:** Pair programming for lab sessions
- **Rural Scaffolding:** Regional metaphors (grain sorting, bus routes, hostel inventories)

Lesson Plan Structure (40 Minutes)

Time	Phase	Teacher Activity	Student Activity
0-5 min	Engage	Present real-world scenario using	Discuss, share observations
5-15 min	Explore	regional metaphor Guide hands-on exploration in	Work in Driver-Navigator pairs,
15-25 min	Explain	pairs Live coding demonstration,	explore software Follow along, ask questions, take
25-35 min	Elaborate	introduce concepts Assign extension activity or	notes Apply learning, solve problems
35-40 min	Evaluate	Parson's problem Conduct formative assessment, assign homework	independently Complete exit ticket, receive dormitory task

Assessment Framework

Assessment Components (Per Class)

Assessment Type	Count	Marks Each	Total
Formative (FA-1 to FA-4)	4	30	120
Unit Tests	4	10	40
Summative (SA-1, SA-2)	2	40	80
Practical Lab Exams	2	20	40
Grand Total	12		280

Resources Required

Hardware

- Computer lab with 1 PC per 2 students (minimum)
- Internet connection (2 Mbps minimum)
- Projector for demonstrations
- Speakers and headphones

Software (Free/Open Source)

- Operating System: Ubuntu Linux or Windows
- Office Suite: LibreOffice / Google Workspace
- Graphics: GIMP, Tux Paint
- Audio: Audacity
- Programming: Scratch
- Browser: Firefox / Chromium

Dormitory Tasks (Zero Lab Access)

For residential schools with limited lab time, these paper-and-pencil tasks can be assigned for supervised evening study:

- **Code Dry-Running:** Trace Scratch programs on paper
- **Variable State Tables:** Track variable values through program execution
- **Flowchart Drawing:** Create flowcharts for algorithms
- **Binary Conversion:** Practice decimal to binary conversions
- **Keyboard Layout:** Practice typing without computer
- **Software Matching:** Match software to their purposes

Capstone Projects

Project 1: School Environment Monitor

Integration: Science + ICT

Description: Students create a Scratch program that simulates monitoring temperature, humidity, and weather conditions in the school campus.

Skills: Data representation, variables, loops, conditionals

Project 2: Hostel Store Inventory

Integration: Mathematics + ICT

Description: Students create a spreadsheet to track hostel store items (rice, dal, sugar) with quantity, rate, and total calculations.

Skills: Spreadsheet formulas, data entry, basic calculations